

Section 1.1: Systems of Linear Equations

Section 1.2: Row Reduction and Echelon Forms

Section 1.3: Vector Equations

Vector

Vectors in $\mathbb{R}^2$

- A matrix with only one column is called a **column vector**, or simply a **vector**.
- An example of a vector with two entries is

$$\mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix},$$

where w_1 and w_2 are any real numbers.

- The set of all vectors with 2 entries is denoted by $\mathbb{R}^2$ (read “r-two”).

Vector

- **Example:** Given $\mathbf{u} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 2 \\ -5 \end{bmatrix}$, find $4\mathbf{u}$, $(-3)\mathbf{v}$, and $4\mathbf{u} + (-3)\mathbf{v}$.

Solution:

$$4\mathbf{u} = \begin{bmatrix} 4 \\ -8 \end{bmatrix}, \quad (-3)\mathbf{v} = \begin{bmatrix} -6 \\ 15 \end{bmatrix} \text{ and}$$
$$4\mathbf{u} + (-3)\mathbf{v} = \begin{bmatrix} 4 \\ -8 \end{bmatrix} + \begin{bmatrix} -6 \\ 15 \end{bmatrix} = \begin{bmatrix} -2 \\ 7 \end{bmatrix}$$

Geometric Descriptions of $\mathbb{R}^2$

- Consider a rectangular coordinate system in the plane. Because each point in the plane is determined by an ordered pair of numbers, we can identify a geometric point (a, b) with the column vector $\begin{bmatrix} a \\ b \end{bmatrix}$.
- So we may regard $\mathbb{R}^2$ as the set of all points in the plane.

Geometric Descriptions of $\mathbb{R}^2$

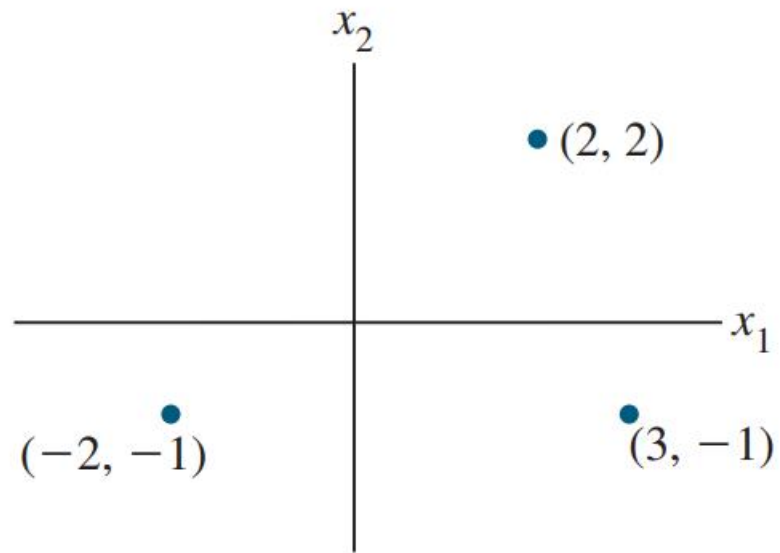


FIGURE 1 Vectors as points.

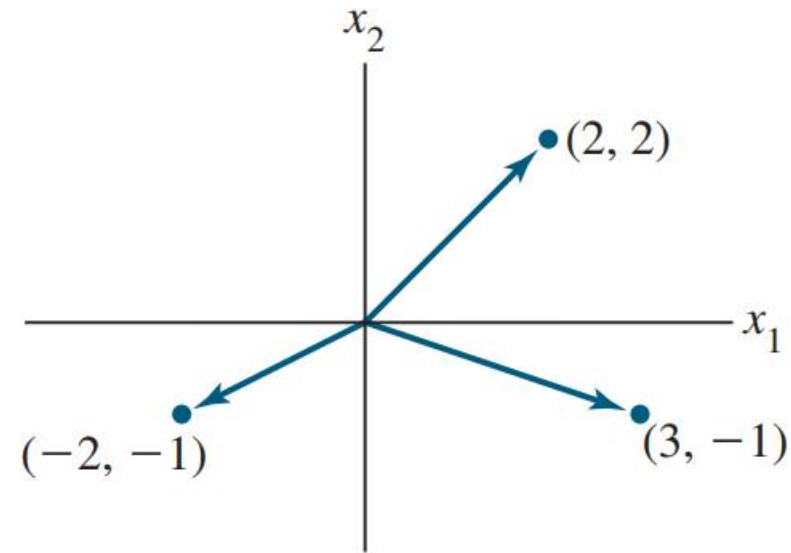
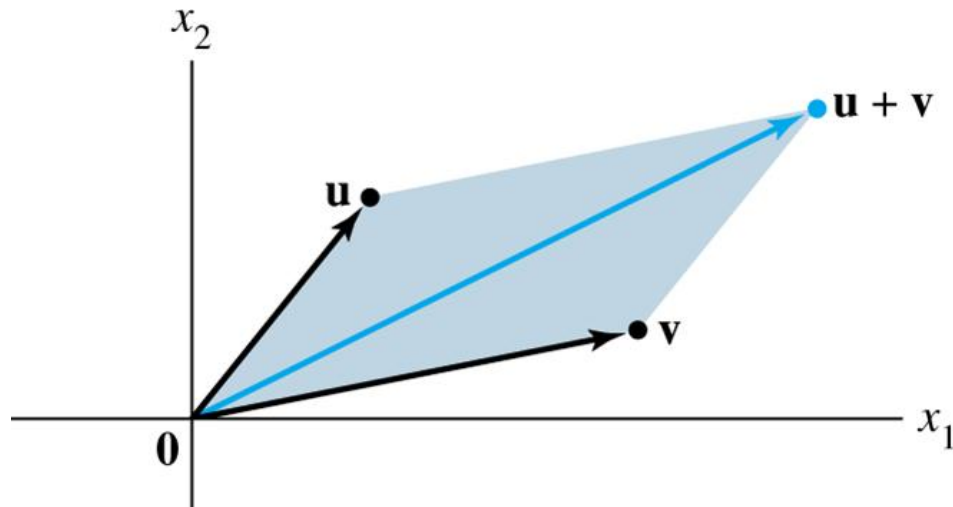


FIGURE 2 Vectors with arrows.

Parallelogram Rule For

- If $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^2$ are represented as points in the plane, then $\mathbf{u} + \mathbf{v}$ corresponds to the fourth vertex of the parallelogram whose other vertices are $\mathbf{u}$, $\mathbf{0}$, and $\mathbf{v}$. See the figure below.



Vectors In Double Lined R Cube and Double Lined R to the n Power

- Vectors in $\mathbb{R}^3$ are 3×1 column matrices with three entries.
- They are represented geometrically by points in a three-dimensional coordinate space, with arrows from the origin.
- If n is a positive integer, $\mathbb{R}^n$ (read “r-n”) denotes the collection of all lists (or **ordered n -tuples**) of n real numbers, usually written as $n \times 1$ column matrices, such as

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}.$$

Algebraic Properties Of Double Lined R to the n Power

- The vector whose entries are all zero is called the **zero vector** and is denoted by **0**.
- For all **u**, **v**, **w** in $\mathbb{R}^n$ and all scalars **c** and **d**:

$$(i) \quad \mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$$

$$(ii) \quad (\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$$

$$(iii) \quad \mathbf{u} + \mathbf{0} = \mathbf{0} + \mathbf{u} = \mathbf{u}$$

$$(iv) \quad \mathbf{u} + (-\mathbf{u}) = \mathbf{0} \text{ where } -\mathbf{u} \text{ denotes } (-1)\mathbf{u}$$

$$(v) \quad c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$$

$$(vi) \quad (c+d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$$

$$(viii) \quad 1\mathbf{u} = \mathbf{u}$$

Linear Combinations

$$(vii) \quad c(d\mathbf{u}) = (cd)(\mathbf{u})$$

$$(viii) \quad 1\mathbf{u} = \mathbf{u}$$

- Given vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$ in $\mathbb{R}^n$ and given scalars $c_1, c_2, \dots, c_p$, the vector $\mathbf{y}$ defined by

$$\mathbf{y} = c_1\mathbf{v}_1 + \dots + c_p\mathbf{v}_p$$

is called a **linear combination** of $\mathbf{v}_1, \dots, \mathbf{v}_p$ with **weights** $c_1, \dots, c_p$.

- The weights in a linear combination can be any real numbers, including zero.

Linear Combinations

- Example: Let $\mathbf{a}_1 = \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix}$, $\mathbf{a}_2 = \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$.

Determine whether $\mathbf{b}$ can be generated (or written) as a linear combination of $\mathbf{a}_1$ and $\mathbf{a}_2$. That is, determine whether weights x_1 and x_2 exist such that

$$x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 = \mathbf{b} \quad \text{--- (1)}$$

If vector equation (1) has a solution, find it.

Linear Combinations

Solution: Use the definitions of scalar multiplication and vector addition to rewrite the vector equation

$$x_1 \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix},$$

$\uparrow \quad \quad \uparrow \quad \quad \uparrow$
 $\mathbf{a}_1 \quad \quad \mathbf{a}_2 \quad \quad \mathbf{b}$

which is same as

$$\begin{bmatrix} x_1 \\ -2x_1 \\ -5x_1 \end{bmatrix} + \begin{bmatrix} 2x_2 \\ 5x_2 \\ 6x_2 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}.$$

Linear Combinations

$$\text{and } \begin{bmatrix} x_1 + 2x_2 \\ -2x_1 + 5x_2 \\ -5x_1 + 6x_2 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}. \quad \text{---- (2)}$$

make the vector equation (1) true if and only if x_1 and x_2 satisfy the following system.

$$\begin{aligned} x_1 + 2x_2 &= 7 \\ -2x_1 + 5x_2 &= 4 \\ -5x_1 + 6x_2 &= -3 \end{aligned} \quad \text{---- (3)}$$

Linear Combinations

- To solve this system, row reduce the augmented matrix of the system as follows.

$$\begin{bmatrix} 1 & 2 & 7 \\ -2 & 5 & 4 \\ -5 & 6 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 7 \\ 0 & 9 & 18 \\ 0 & 16 & 32 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 7 \\ 0 & 1 & 2 \\ 0 & 16 & 32 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

- The solution of (3) is $x_1 = 3$ and $x_2 = 2$. Hence **b** is a linear combination of **a**₁ and **a**₂, with weights $x_1 = 3$ and $x_2 = 2$. That is,

$$3 \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix} .$$

Linear Combinations

- **A vector equation**

$$x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 + \dots + x_n \mathbf{a}_n = \mathbf{b}$$

has the same solution set as the linear system whose augmented matrix is

$$[\mathbf{a}_1 \quad \mathbf{a}_2 \quad \dots \quad \mathbf{a}_n \quad \mathbf{b}].$$

In particular, $\mathbf{b}$ can be generated by a linear combination of $\mathbf{a}_1, \dots, \mathbf{a}_n$ if and only if there exists a solution to the linear system corresponding to the matrix.

Linear Combinations

- **Definition:** If $\mathbf{v}_1, \dots, \mathbf{v}_p$ are in $\mathbb{R}^n$, then the set of all linear combinations of $\mathbf{v}_1, \dots, \mathbf{v}_p$ is denoted by $\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ and is called the **subset of $\mathbb{R}^n$ spanned** (or **generated**) by $\mathbf{v}_1, \dots, \mathbf{v}_p$. That is, $\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ is the collection of all vectors that can be written in the form

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_p\mathbf{v}_p$$

with $c_1, \dots, c_p$ scalars.

Section 1.4: The Matrix Equation $Ax = b$

Section 1.5: Solution Sets of Linear Systems

Section 1.7: Linear Independence

Matrix Equation $Ax = b$

- **Definition:** If A is an $m \times n$ matrix, with columns $a_1, \dots, a_n$, and if x is in $\mathbb{R}^n$, then the **product** of A and x , denoted by Ax , is the linear combination of the columns of A using the corresponding entries in x as weights; that is,

$$Ax = [a_1 \quad a_2 \quad \cdots \quad a_n] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1 a_1 + x_2 a_2 + \cdots + x_n a_n.$$

- Ax is defined only if the number of columns of A equals the number of entries in x .

Matrix Equation $Ax = b$

- Example:

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & -5 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \\ 7 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} 2 \\ -5 \end{bmatrix} + 7 \begin{bmatrix} -1 \\ 3 \end{bmatrix}$$

Matrix Equation $Ax = b$

- **Example:** For v_1, v_2, v_3 in $\mathbb{R}^m$, write the linear combination $3v_1 - 5v_2 + 7v_3$ as a matrix times a vector.
- **Solution:** Place v_1, v_2, v_3 into the columns of a matrix A and place the weights $3, -5,$ and 7 into a vector x .
- That is,

$$3v_1 - 5v_2 + 7v_3 = [v_1 \quad v_2 \quad v_3] \begin{bmatrix} 3 \\ -5 \\ 7 \end{bmatrix} = Ax$$

Matrix Equation $Ax = b$

- Now, write the system of linear equations as a vector equation involving a linear combination of vectors.
- For example, the following system

$$\begin{aligned}x_1 + 2x_2 - x_3 &= 4 \\ -5x_2 + 3x_3 &= 1\end{aligned}\quad \text{---- (1)}$$

is equivalent to

$$\begin{aligned}x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ -5 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 3 \end{bmatrix} \\ = \begin{bmatrix} 4 \\ 1 \end{bmatrix} . \quad \text{---- (2)}\end{aligned}$$

Matrix Equation $Ax = b$

- As in the given example, the linear combination on the left side is a matrix times a vector, so that (2) becomes

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & -5 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}. \quad \text{--- (3)}$$

- Equation (3) has the form $Ax = b$. Such an equation is called a **matrix equation**.

Matrix Equation $Ax = b$

- **Theorem:** If A is an $m \times n$ matrix, with columns $a_1, \dots, a_n$, and if b is in $\mathbb{R}^m$, then the matrix equation

$$Ax = b$$

has the same solution set as the vector equation

$$x_1 a_1 + x_2 a_2 + \dots + x_n a_n$$

which, in turn, ~~has~~ $= b$ the same solution set as the system of linear equations whose augmented matrix is

$$[a_1 \quad a_2 \quad \cdots \quad a_n \quad b].$$

Existence of Solutions

- The equation $A\mathbf{x} = \mathbf{b}$ has a solution if and only if $\mathbf{b}$ is a linear combination of the columns of A .
- **Theorem:** Let A be an $m \times n$ matrix. Then the following statements are logically equivalent. That is, for a particular A , either they are all true statements or they are all false.
 - a. For each $\mathbf{b}$ in $\mathbb{R}^m$, the equation $A\mathbf{x} = \mathbf{b}$ has a solution.
 - b. Each $\mathbf{b}$ in $\mathbb{R}^m$ is a linear combination of the columns of A .
 - c. The columns of A span $\mathbb{R}^m$.
 - d. A has a pivot position in every row.

Computation of $A\mathbf{x}$

- **Example:** Compute $A\mathbf{x}$, where

$$A = \begin{bmatrix} 2 & 3 & 4 \\ -1 & 5 & -3 \\ 6 & -2 & 8 \end{bmatrix} \text{ and } \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}.$$

- **Solution:** From the definition,

$$\begin{aligned} & \begin{bmatrix} 2 & 3 & 4 \\ -1 & 5 & -3 \\ 6 & -2 & 8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \\ &= x_1 \begin{bmatrix} 2 \\ -1 \\ 6 \end{bmatrix} + x_2 \begin{bmatrix} 3 \\ 5 \\ -2 \end{bmatrix} + x_3 \begin{bmatrix} 4 \\ -3 \\ 8 \end{bmatrix} \end{aligned}$$

Computation of Ax

$$= \begin{bmatrix} 2x_1 \\ -x_1 \\ 6x_1 \end{bmatrix} + \begin{bmatrix} 3x_2 \\ 5x_2 \\ -2x_2 \end{bmatrix} + \begin{bmatrix} 4x_3 \\ -3x_3 \\ 8x_3 \end{bmatrix} \quad \text{--- (1)}$$

$$= \begin{bmatrix} 2x_1 + 3x_2 + 4x_3 \\ -x_1 + 5x_2 - 3x_3 \\ 6x_1 - 2x_2 + 8x_3 \end{bmatrix}.$$

Computation of $A\mathbf{x}$

- Example:

Let $A = \begin{bmatrix} 1 & 3 & 4 \\ -4 & 2 & -6 \\ -3 & -2 & -7 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$. Is the equation $A\mathbf{x} = \mathbf{b}$

consistent for all possible b_1, b_2, b_3 ?

Solution:

$$\begin{aligned} \begin{bmatrix} 1 & 3 & 4 & b_1 \\ -4 & 2 & -6 & b_2 \\ -3 & -2 & -7 & b_3 \end{bmatrix} &\sim \begin{bmatrix} 1 & 3 & 4 & b_1 \\ 0 & 14 & 10 & b_2 + 4b_1 \\ 0 & 7 & 5 & b_3 + 3b_1 \end{bmatrix} \\ &\sim \begin{bmatrix} 1 & 3 & 4 & b_1 \\ 0 & 14 & 10 & b_2 + 4b_1 \\ 0 & 0 & 0 & b_3 + 3b_1 - \frac{1}{2}(b_2 + 4b_1) \end{bmatrix} \end{aligned}$$

The equation $A\mathbf{x} = \mathbf{b}$ is *not* consistent for every $\mathbf{b}$

because some choices of $\mathbf{b}$ can make $b_1 - \frac{1}{2}b_2 + b_3$ nonzero.

Row—vector Rule for Computing Ax

- The matrix with 1s on the diagonal and 0s elsewhere is called an **identity matrix** and is denoted by I .

- For example, $I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is an identity matrix.

If $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, compute $I\mathbf{x}$.

Properties of the Matrix–vector Product $A\mathbf{x}$

- **Theorem:** If A is an $m \times n$ matrix, $\mathbf{u}$ and $\mathbf{v}$ are vectors in $\mathbb{R}^n$, and c is a scalar, then
 - a. $A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v}$,
 - b. $A(c\mathbf{u}) = c(A\mathbf{u})$.